

Problem 1.15

The process of finding the present value P of an amount A , due at the end of t years, is called discounting A . The difference $A - P$ is called the discount on A . Notice that the discount on A is also the interest on P . For example, if \$1150 is the discounted value of \$1250, due at the end of 7 months, the discount on the \$1250 is \$100. What is the interest on \$1150 for the same period of time?

Solution

The discount on an amount A is defined as

$$A - P.$$

Here,

$$A = 1250, \quad P = 1150.$$

So the discount is

$$1250 - 1150 = 100.$$

Since the discount on A is also the interest on P , the interest on \$1150 for the same period of time is

$$\boxed{\$100}.$$